

Daniel Pan

Software Engineer | Full-Stack, Systems & AI

Florida, USA | U.S. Citizen | Open to Relocation

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TECHNICAL SKILLS

Languages: C++, Python, Java, C#, JavaScript, TypeScript, SQL, Objective-C

Frontend: React, Next.js, Angular, NgRx, HTML, CSS, Tailwind CSS

Backend & Cloud: REST APIs, Next.js API Routes, SQLite, AWS S3, JWT Authentication, CSRF Protection

Tools & Frameworks: Git, Cypress, Gradle, IntelliJ, CLion, VS Code, Theos, Logos, FFmpeg

PROFESSIONAL EXPERIENCE

Airline Tariff Publishing Company (ATPCO) | Angular, Cypress

JUL 2024 – DEC 2024

Software Engineer Intern

Dulles, VA

- Developed Angular frontend features for airline pricing applications, integrating REST APIs with NgRx state management and cached API responses.
- Built reactive UI workflows that maintained consistent client-side state and improved reliability of customer-facing application features.
- Created, refactored, and maintained Cypress end-to-end tests for evolving data workflows, UI components, and release-ready functionality.

Museum of Discovery and Science | Unity, C#

NOV 2019 – MAY 2021

Software Engineer Intern

Fort Lauderdale, FL

- Developed Unity-based augmented reality exhibit software integrating 3D models, animations, calibration work, and interactive visitor-facing behaviors.
- Collaborated with museum directors and project stakeholders to iterate on exhibit behavior based on visitor feedback and improve the educational experience.

PROJECTS

Self-Hosted Collaborative Storage Platform | React, Next.js, SQLite, AWS S3, Tailwind CSS

2024 – Present

- Built an open-source cloud storage application with authentication, folder hierarchy management, tagging, file previews, multi-file upload, and deletion workflows.
- Developed Next.js API routes backed by SQLite and AWS S3 for metadata persistence, private object storage, secure media access, and file operations.
- Implemented JWT authentication, CSRF protection, responsive React interfaces, reusable UI components, and server-sent progress updates for upload feedback.

Video Frame Analysis Pipeline | Python, scikit-learn, FFmpeg, Whisper

2026 – Present

- Built a Python pipeline that analyzes League of Legends gameplay videos, detects HUD state changes, reconstructs ability and item usage, and generates structured timelines, overlays, and review artifacts.
- Trained per-region HUD classifiers from labeled video crops using image features and scikit-learn models to classify ready, cooldown, disabled, missing, and recast states.
- Integrated FFmpeg, Whisper transcription, glossary-driven translation, contextual frame extraction, and local LLM workflows for Chinese-to-English subtitle review.

iOS Tweak for Mobile Apple Devices | Objective-C, Logos, Theos

2021 – 2023

- Developed and published an installable iOS tweak for jailbroken devices that modified stock iOS behavior and added dynamic wallpaper cycling across iOS 12-15.
- Used Theos, Logos, Objective-C hooks, iOS framework interfaces, and framework-level class modifications to integrate with native system functionality.
- Improved rapid wallpaper swapping through event handling and memory/cache management, then published a package repository reaching 1,700+ downloads.

Minesweeper Machine Learning Environment | C++, Raylib, multithreading

2025 – Present

- Built C++ Minesweeper engines with separate rendering, game-state, and solver architectures to support both player-facing gameplay and algorithm experimentation.
- Implemented multithreaded computation to decouple solver execution from visualization, making algorithm steps easier to inspect while preserving responsive rendering.
- Added viewport culling and large-board support for generated boards up to 10,000x10,000, emphasizing performance, algorithms, and scalable simulation design.

Minecraft Server-Side Generation and Scaling Mods | Java, Neoforge, Forge, Gradle

2026 – Present

- Developed configurable Minecraft server systems spanning runtime world generation, persistent state management, entity scaling, and datapack-driven configuration.
- Built a mob scaling framework using statistical feedback-control concepts to dynamically adjust difficulty based on player progression.
- Refactored shared logic across NeoForge and Forge APIs while preserving platform-specific behavior, configuration flexibility, and third-party mod compatibility.

EDUCATION

Georgia Institute of Technology

May 2022 – Dec 2025

Bachelor of Science in Computer Science

Atlanta, GA

Relevant coursework: Machine Learning (CS 4641), Artificial Intelligence (CS 3600), Database Systems (CS 4400), Design and Analysis of Algorithms (CS 3510), Data Structures & Algorithms (CS 1332), Computer Organization & Programming (CS 2110), Objects and Design (CS 2340)